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10 CONCEPTUAL Q&A ROBOTICS · PRACTICE WORKSHEET

CONCEPTUAL SERIES

NVIDIA Isaac

Name: ______________________ Cohort/Batch: ______________________ Date: ______________ Score: __________

NVIDIA ISAAC PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Robotics

TOTAL: 10 QUESTIONS

Q1 What is NVIDIA Isaac and what problem in Robotics does it solve? *Threat Model*

Q2 What are the core components or concepts in NVIDIA Isaac? *Architecture*

Q3 How does NVIDIA Isaac compare to its main alternatives? *Configuration*

Q4 How do you get started with NVIDIA Isaac for a new project? *Hardening*

Q5 What configuration options most affect NVIDIA Isaac's behavior in production? *SSO / IdP*

Q6 How does NVIDIA Isaac handle reliability and high availability? *Observability*

Q7 What observability does NVIDIA Isaac provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in NVIDIA Isaac? *Compliance*

Q9 What security best practices apply when running NVIDIA Isaac? *Testing*

Q10 What mistakes do teams commonly make when adopting NVIDIA Isaac? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 NVIDIA Isaac is a tool or platform

Mitigation

NVIDIA Isaac is a tool or platform that automates or simplifies a specific concern in the Robotics domain, reducing manual effort and operational complexity for engineering teams.

A2 NVIDIA Isaac has key building blocks that

Primitives

NVIDIA Isaac has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 NVIDIA Isaac trades off simplicity against feature

Hardening

NVIDIA Isaac trades off simplicity against feature richness differently than its alternatives. Choose NVIDIA Isaac when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install NVIDIA Isaac via its package manager

Gotchas

Install NVIDIA Isaac via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include concurrency

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 NVIDIA Isaac provides HA through leader election,

Observability

NVIDIA Isaac provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 NVIDIA Isaac exposes metrics (Prometheus endpoint or

Automation

NVIDIA Isaac exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from misconfigured

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run NVIDIA Isaac with the principle of

Assessment

Run NVIDIA Isaac with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the documentation and

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.